

Multiple Pattern Search

Topic.

String algorithms, Aho–Corasick automaton, multiple pattern matching.

Statement.

Given a set of patterns T and a set of strings S .

For each string from S , check whether it contains at least one string from T as a substring.

Input.

The first line contains an integer

$$n,$$

the number of strings in the set T .

Each of the next n lines contains a non-empty pattern string of length at most 80.

The remaining part of the input contains strings from the set S .

Each string from S consists of ASCII characters with codes from 32 to 126 inclusive.

A string from S may be empty and has length at most 250.

The input file size does not exceed 1 MB.

Output.

Print all strings from the set S that contain at least one string from the set T as a substring.

The strings must be printed in the same order as they appear in the input.

Implementation.

The implementation is provided separately in the file

`solution.cpp`.